# Static random hashes: exact first-collision law

**Result.** Lean proves that an arbitrary finite state update driven by independently sampled static vertex labels has exactly the same first-collision survival probabilities as a chronological iid label walk stopped when a state repeats. The proof uses actual deterministic paths and actual product PMFs, without a survival/mixing/collision conclusion supplied as a premise. [pcc:unverified]

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|---|---|
| Actual models | Static labels: one independently sampled label per state<br>Chronological labels: independent choices at each time |
| First collision | Survival probabilities agree at every finite time<br>Actual E[first repeat] = sum of iid self-avoidance tails |
| Boundary control | Z3, steps1,2: word0,1,1 revisits0 with conflicting labels<br>Static word mass0; iid mass1/8; unrestricted law false |

**Independent queries are derived.** A prescribed finite word determines an actual state path. Lean proves that a static hash produces this word exactly when its labels agree at the corresponding queried states. If the path is self-avoiding, these states are distinct. An arbitrary finite injective-query cylinder theorem derives the word probability as the product of label weights, using actual coordinate gates, products of sums and extension by one. Independence is not asserted for a repeated state. [pcc:unverified]

**Exact survival and expectation.** Decomposing the survival event into disjoint chronological word events gives equality with the actual iid self-avoidance mass at every time, for all finite state/label types and normalized nonnegative weights. On every nonzero modulus p, Lean links the unchanged actual Nat.find least-repeat definition to prefix injectivity. Its tail probability is the iid self-avoidance mass; its expectation is the finite sum of those tails for times0 through p-1. Any state rule and starting state are allowed, so adding walks are included. [pcc:unverified]

**Direct disproof beyond the stopping boundary.** The actual prime-order group Z3 has distinct nonzero increments1,2 and strictly positive uniform labels. Word0,1,1 would move0→1→0, then require label1 at state0 although its first label was0. Lean proves this word impossible under every static hash, with static mass0 and iid mass1/8, and directly negates the unrestricted chronological word-mass identity. Kernel-checked finite decisions supply the small control. [pcc:unverified]

**Boundary and prior work.** This closes a first-collision model bridge. It does not prove the limiting random-increment-table penalty, practical fixed/keyedhash refinement or compiler execution. Bernstein–Lange2013 already identifies higher-degree local anticollisions beyond the degree-one factor; those asymptotic heuristic corrections are credited and are not proved here. A finite counterexample is not presented as an asymptotic refutation. [pcc:unverified]

**Checks.** Whole new-project gate37/0 failures/0 trusted native, only propext/Classical.choice/Quot.sound. Complete1421-character bindingbe9c6e9f61b3fd7984f70666f38565d4b55df432cbeecdc75723d02c47859a1f, independent native1complete/0prefix/0unresolved; procedural parser/hash evidence, not kernel refinement. Eight complete build snapshots including six failures: 97files/41objects/14148-byte lossless XZ, plus final gate files. [pcc:unverified]

**Sources and assurance.** Galbraithch14v2 full selected printed288/296/297 (PDF4/12/13) and Bernstein–Lange Two grumpy giants and a baby full selected94–97 (PDF9–12) read/visually inspected. No fullpaper-read or novelty claim. Prior actual hash-PMF/moment and least-repeat proofs reused unchanged; official Mathlib finite products/extension/iteration/find APIs checked. KB/Conductor unavailable, primary/prior/official fallback. Lean4.35.0/Mathlib56a00585…, defaults, no holes/custom axioms/native evaluator/global flags/raised limits/waivers/new crypto experiment/spend/upload. Signed standing unverified. [pcc:unverified]